



AI CODE ANALYZER

FULL-STACK INTELLIGENCE SYSTEM

[STATUS: SYSTEM ONLINE // MODEL: DEEPSEEK-CODER]

| PROBLEM STATEMENT

- ⚠️ **Manual Fatigue:** Peer reviews are slow, inconsistent, and often miss critical logic bugs.
- 🧠 **Data Privacy:** Sending proprietary code to cloud LLMs (OpenAI) risks IP leakage.
- 💰 **Cost Barriers:** High token usage fees for enterprise-scale codebases.
- ⚙️ **Latency:** Cloud-based analyzers introduce significant wait times during dev cycles.



THE MISSION



Eliminate Technical Debt: Real-time semantic analysis to catch architectural flaws.



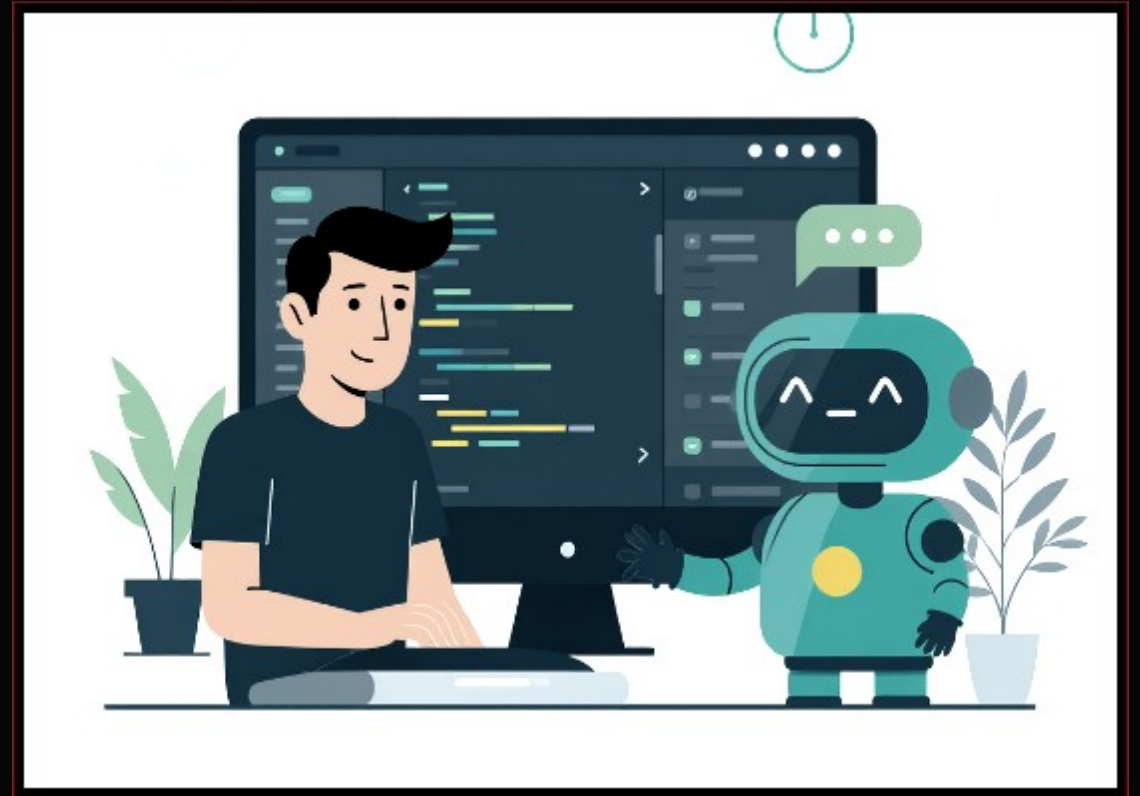
Security Hardening: Automatic detection of vulnerabilities in 15+ languages.



Vision-Powered: Extract and analyze code directly from IDE screenshots.



Decision Support: AI-driven "Should Push" metrics for Git workflows.



SYSTEM NUCLEUS



VITE FRONTEND

High-speed React interface with
Black/Red Cyberpunk UI & Radar Charts.



FASTAPI BACKEND

Async Python engine handling Risk
Scoring, SQLite logging, and AI
orchestration.



OLLAMA ENGINE

Local LLM hosting: DeepSeek-Coder-6.7B
& Qwen2.5-Coder-7B.

OS: macOS 13+ (Silicon Optimized) // DB: SQLite // VISION: LLaVA

DATA TRAJECTORY



| PROJECT WORKING FLOW

1. Ingest

User pastes code or uploads a screenshot (OCR parsing).

2. Process

FastAPI constructs a high-context semantic prompt.

3. Infer

Ollama runs DeepSeek-Coder to find smells & logic bugs.

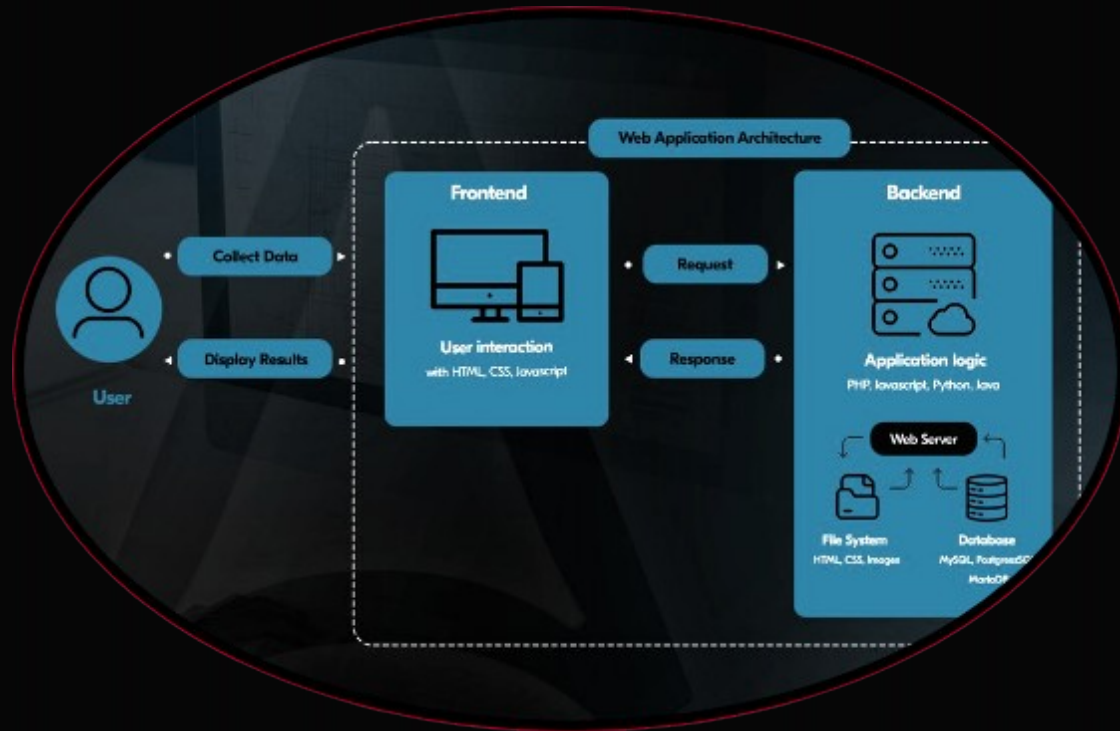
4. Verify

Risk Engine generates 0-100 score and diff viewer.

5. Push

Verified code is deployed directly to GitHub.

METRIC DASHBOARD



- **Risk Score (0-100):** Classifies code from Safe to Critical.

- **Radar Charts:** 5-point analysis (Security, Performance, etc.)

- **Semantic Diff:** Visual Red/Green side-by-side refactoring.

- **History Replay:** Track risk trends across all sessions.

OLLAMA INTEGRATION

MODEL HIERARCHY

```
> ollama pull deepseek-coder
```

Target: 6.7B Model - Optimized for macOS performance.

```
> ollama pull qwen2.5-coder
```

Target: 7B Model - Multilingual semantic expert.

```
> ollama pull llava
```

Target: Vision Support - Screenshots extraction.

🔌 **Offline Fallback:** Static analysis triggers if Ollama server is unreachable.

🍏 **Apple Silicon:** Leverages Metal acceleration for instant code feedback.

FUTURE UPGRADES

PHASE 1

Direct IDE Integration (VS Code /
JetBrains Plugins)

PHASE 2

Multi-modal Chat Interface for
deeper refactor Q&A

PHASE 3

Team Collaboration & Shared SQLite
Cloud Sync

| COMPANY VALUE

Metric	Impact	Result
Review Speed	85% Faster	Increased Feature Velocity
Security	Proactive Fixes	Reduced Patch Cost
Data Privacy	100% Local	Compliance Verified

TERMINATED_01

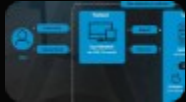
THANK YOU

IMAGE SOURCES



https://www.softwareseni.com/wp-content/uploads/2025/08/header_image_CSW202508111258.png

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Source: acropolium.com